

## OSTROV PREOBRAZENIJ, Russian Federation

WMO: 215040

Lat: 74.667N

Lon: 112.933E

Elev: 187

StdP: 14.60

Time Zone: 9.00 (E09)

Period: 82-96

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                      | (o)  | (p)   |
| 1             | -42.1      | -38.5 | -45.5                         | 0.4 | -41.8 | -41.8 | 0.5 | -38.2 | 37.8                  | -11.5 | 30.8 | -14.9 | 8.1                      | 180  | 1.215 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.3                    | 53.8            | 49.7 | 48.7 | 45.8 | 44.8 | 42.5 | 50.1                | 53.6 | 45.7 | 48.4 | 42.8 | 44.7 | 12.5                 | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 47.7                            | 49.3 | 52.1 | 43.7 | 42.3 | 47.0 | 41.0 | 38.1 | 43.9 | 20.5          | 53.6 | 18.1 | 48.8 | 16.7 | 44.8 | 69.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 34.3 | 29.7 | 26.7                       | DB  | -44.0              | 65.2 | 4.0                                                | 6.3 | -46.9      | 69.7 | -49.2      | 73.3 | -51.5      | 76.9 | -54.4 | 81.4 |
| (5)               |      |      |                            | WB  | -44.1              | 58.1 | 3.9                                                | 5.5 | -46.9      | 62.0 | -49.2      | 65.3 | -51.4      | 68.4 | -54.2 | 72.4 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep  | Oct   | Nov  | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)  | (n)   | (o)  | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 6.4    | -22.1 | -19.9 | -16.1 | -4.4  | 14.7  | 30.9  | 38.1  | 37.0  | 30.3 | 11.6  | -9.7 | -14.9 |
|                                                                               | DBStd       | 24.04  | 10.63 | 10.18 | 10.12 | 9.87  | 7.10  | 6.43  | 5.32  | 5.65  | 5.67 | 10.66 | 8.20 | 10.37 |
|                                                                               | HDD50       | 15906  | 2235  | 1958  | 2049  | 1631  | 1094  | 574   | 375   | 405   | 591  | 1191  | 1790 | 2013  |
|                                                                               | HDD65       | 21372  | 2700  | 2378  | 2514  | 2081  | 1559  | 1023  | 835   | 867   | 1041 | 1656  | 2240 | 2478  |
|                                                                               | CDD50       | 9      | 0     | 0     | 0     | 0     | 0     | 1     | 5     | 3     | 0    | 0     | 0    | 0     |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0    | 0     |
|                                                                               | CDH74       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0    | 0     | 0    | 0     |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0     | 0    | 0     |
| Wind                                                                          | WSAvg       | 13.0   | 10.9  | 11.9  | 12.2  | 14.2  | 13.9  | 13.6  | 13.2  | 14.2  | 13.5 | 14.2  | 12.4 | 11.8  |
| Precipitation                                                                 | PrecAvg     | 11.1   | 0.7   | 0.5   | 0.6   | 0.6   | 0.9   | 0.8   | 1.2   | 1.6   | 1.4  | 1.3   | 0.9  | 0.7   |
|                                                                               | PrecMax     | 15.6   | 2.1   | 1.1   | 1.0   | 1.6   | 1.9   | 1.7   | 2.5   | 3.1   | 3.4  | 2.3   | 1.7  | 1.5   |
|                                                                               | PrecMin     | 6.4    | 0.1   | 0.0   | 0.1   | 0.0   | 0.2   | 0.2   | 0.4   | 0.6   | 0.4  | 0.4   | 0.3  | 0.3   |
|                                                                               | PrecStd     | 2.0    | 0.4   | 0.2   | 0.2   | 0.4   | 0.4   | 0.4   | 0.5   | 0.7   | 0.7  | 0.4   | 0.4  | 0.3   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 9.1   | 12.7  | 17.4  | 23.2  | 32.2  | 54.6  | 67.9  | 58.7  | 45.9 | 32.1  | 15.1 | 16.6  |
|                                                                               |             | MCWB   | 8.4   | 12.0  | 15.7  | 21.8  | 30.9  | 48.7  | 60.9  | 53.5  | 43.8 | 31.8  | 14.2 | 16.3  |
|                                                                               | 2%          | DB     | 3.5   | 3.4   | 8.7   | 17.4  | 29.9  | 46.9  | 55.5  | 53.1  | 42.0 | 29.8  | 10.4 | 7.3   |
|                                                                               |             | MCWB   | 2.8   | 2.8   | 8.2   | 16.5  | 29.1  | 43.7  | 50.9  | 49.5  | 41.2 | 29.0  | 9.6  | 6.8   |
|                                                                               | 5%          | DB     | -1.2  | -0.9  | 3.1   | 13.3  | 27.5  | 40.8  | 49.6  | 48.3  | 39.4 | 28.0  | 5.8  | 2.9   |
|                                                                               |             | MCWB   | -1.7  | -1.5  | 2.7   | 12.5  | 26.5  | 38.4  | 46.5  | 45.7  | 38.6 | 27.1  | 5.3  | 2.4   |
|                                                                               | 10%         | DB     | -5.9  | -5.2  | -2.0  | 9.1   | 25.0  | 37.6  | 44.9  | 44.8  | 37.3 | 26.3  | 2.0  | -0.4  |
|                                                                               |             | MCWB   | -6.3  | -5.7  | -2.4  | 8.3   | 24.0  | 35.8  | 42.5  | 42.9  | 36.5 | 25.4  | 1.4  | -0.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.5   | 12.2  | 16.1  | 22.1  | 31.2  | 48.4  | 60.2  | 54.0  | 43.7 | 32.0  | 14.4 | 16.2  |
|                                                                               |             | MCDB   | 9.1   | 12.9  | 17.3  | 23.5  | 31.8  | 53.1  | 66.8  | 57.5  | 44.7 | 32.1  | 14.8 | 17.4  |
|                                                                               | 2%          | WB     | 2.8   | 2.6   | 7.9   | 16.7  | 29.2  | 43.3  | 51.1  | 50.0  | 41.4 | 29.4  | 9.8  | 6.9   |
|                                                                               |             | MCDB   | 3.3   | 3.0   | 8.5   | 17.5  | 29.9  | 46.6  | 54.8  | 53.0  | 42.0 | 29.7  | 10.3 | 7.4   |
|                                                                               | 5%          | WB     | -1.7  | -1.7  | 2.5   | 12.7  | 26.7  | 38.5  | 46.4  | 45.9  | 38.8 | 27.3  | 5.2  | 2.5   |
|                                                                               |             | MCDB   | -1.3  | -1.1  | 3.0   | 13.5  | 27.5  | 40.3  | 49.6  | 48.0  | 39.3 | 28.0  | 5.7  | 2.9   |
|                                                                               | 10%         | WB     | -6.4  | -5.8  | -2.5  | 8.5   | 24.1  | 35.9  | 42.9  | 43.1  | 36.6 | 25.3  | 1.5  | -0.9  |
|                                                                               |             | MCDB   | -6.0  | -5.4  | -2.1  | 9.2   | 25.1  | 37.3  | 45.0  | 44.8  | 37.2 | 26.1  | 2.0  | -0.4  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.3   | 9.9   | 10.7  | 10.4  | 7.4   | 6.0   | 8.3   | 6.3   | 4.7  | 7.1   | 7.9  | 9.0   |
|                                                                               |             | MCDBR  | 15.0  | 14.2  | 16.7  | 14.8  | 8.5   | 12.9  | 18.2  | 13.5  | 6.7  | 7.6   | 11.2 | 11.4  |
|                                                                               |             | MCWBR  | 14.7  | 14.2  | 16.6  | 14.6  | 8.0   | 9.7   | 14.0  | 10.3  | 6.1  | 7.9   | 11.1 | 11.2  |
|                                                                               | 5% WB       | MCDBR  | 14.9  | 14.4  | 16.4  | 14.9  | 8.0   | 12.3  | 17.3  | 13.4  | 6.6  | 7.8   | 11.2 | 11.4  |
|                                                                               |             | MCWBR  | 14.6  | 14.3  | 16.3  | 14.5  | 7.7   | 9.6   | 14.1  | 10.6  | 6.2  | 8.1   | 11.1 | 11.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | N/A    | 0.172 | 0.267 | 0.300 | 0.324 | 0.326 | 0.361 | 0.349 | 0.287 | N/A  | N/A   | N/A  | N/A   |
|                                                                               | taud        | N/A    | 1.873 | 2.006 | 2.054 | 2.078 | 2.262 | 2.321 | 2.390 | 2.407 | N/A  | N/A   | N/A  | N/A   |
|                                                                               | Ebn at Noon | 0      | 92    | 196   | 245   | 258   | 266   | 248   | 233   | 207   | N/A  | 0     | 0    | 0     |
|                                                                               | Edh at Noon | 0      | 6     | 22    | 33    | 38    | 33    | 30    | 24    | 16    | N/A  | 0     | 0    | 0     |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 0      | 32    | 434   | 1202  | 1788  | 1954  | 1571  | 886   | 364   | 73   | 0     | 0    | 0     |
|                                                                               | RadStd      | 0      | 4     | 31    | 71    | 91    | 162   | 148   | 104   | 35    | 9    | 0     | 0    | 0     |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page